

Machine Learning Assignment

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GitHub Repository: <https://github.com/prashant-vsg/ML-Assignment-1-IMT2024075>

The repository contains the training, validation, and inference scripts used for both polynomial regression tasks, along with the generated prediction files.

1. Methodology

1.1 Polynomial Regression

Polynomial regression was used to model nonlinear relationships between the input variables and the target y . Polynomial features, including interaction terms, were generated using `PolynomialFeatures` from scikit-learn.

The models were initially fitted using ordinary least squares (OLS), which minimizes the squared prediction error.

1.2 Ridge Regularization

Ridge regression was then tested to reduce overfitting caused by large polynomial coefficients.

Its loss function is:

$$L = \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p w_j^2$$

The regularization strength was tuned by testing different alpha values. `StandardScaler` was applied to the polynomial features before Ridge regression.

1.3 Model Selection

For both datasets, polynomial degrees and feature subsets were evaluated using 5-fold cross-validation.

- Var1: Degrees 1–10, with 63 possible feature subsets.
- Var2: Degrees 1–20, with 7 possible feature subsets.

Ridge regression was then tested across different degrees and regularization strengths. The final candidates were compared across three additional random seeds (7, 21 and 123) to check their stability.

1.4 Evaluation

Mean Squared Error (MSE) was used as the main metric, while R^2 was used to measure how well the model explained the variation in the target.

The models were selected based on cross-validation performance and then trained on the complete training datasets before generating test predictions.

2. Steam Turbine Optimization (Var1)

The dataset contains 1,000 training samples with six input features. Polynomial degrees from 1 to 10 were tested.

Initially, ordinary least squares (OLS) gave the lowest cross-validation MSE at degree 4. Ridge regression was then tested to see if regularization could improve the results.

2.1 Results

Model	Degree	Alpha	Mean CV MSE	Mean CV R^2
OLS	4	—	0.792060	0.921626
Ridge	4	10	0.736474	—
Ridge	5	15	0.552955	0.945370

Ridge regression performed better than OLS, reducing the validation MSE by approximately 30.2%.

All 63 possible feature subsets were tested at degree 5 with $\alpha = 15$. Using all six features gave the lowest error.

2.2 Model Comparison

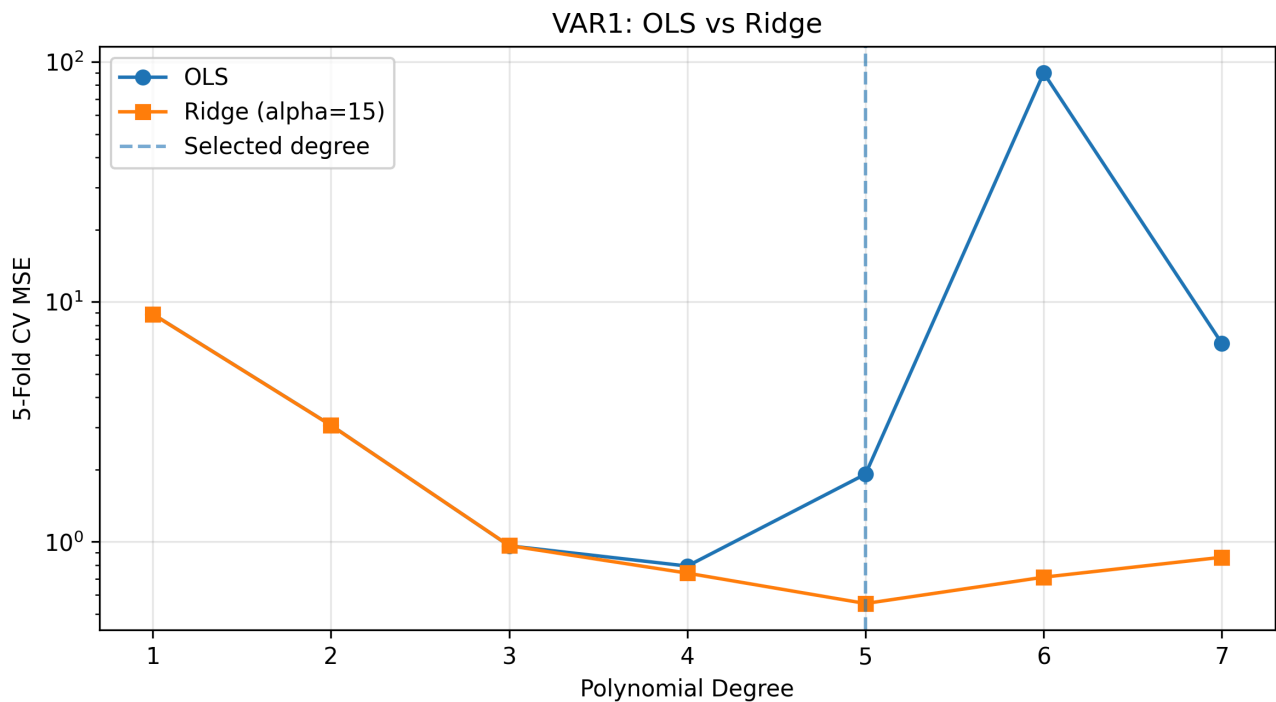


Figure 1: Comparison of OLS and Ridge regression across polynomial degrees using 5-fold cross-validation.

The graph shows that OLS becomes unstable at higher degrees, while Ridge controls the increase in error. Degree 5 gave the lowest MSE for Ridge at the selected regularization strength.

2.3 Final Model

- **Model:** Ridge regression
- **Polynomial degree:** 5
- **Regularization strength:** $\alpha = 15$
- **Selected features:** $x_1, x_2, x_3, x_4, x_5, x_6$

The selected Ridge model was also evaluated using three additional random seeds (7, 21 and 123), giving an average CV MSE of **0.523408**.

3. Subterranean Thermal Reservoir Mapping (Var2)

The dataset contains 1,000 training samples with three input features representing spatial coordinates. Polynomial degrees from 1 to 20 were tested.

OLS gave the lowest cross-validation MSE at degree 8. Ridge regression was then tested to check whether higher-degree models could perform better with regularization.

3.1 Results

Model	Degree	Alpha	Mean CV MSE	Mean CV R ²
OLS	8	—	0.246520	0.994729

Model	Degree	Alpha	Mean CV MSE	Mean CV R ²
Ridge	11	1	0.237452	—
Ridge	12	3	0.234399	—
Ridge	10	0.6	0.236797	0.994949

Although degree 12 gave the lowest MSE on the original validation split, degree 10 was selected after comparing performance across three additional random seeds.

All seven feature subsets were tested using Ridge at degree 12 with $\alpha = 3$. Using all three features gave the lowest validation error, supporting their use in the final model.

3.2 Model Comparison

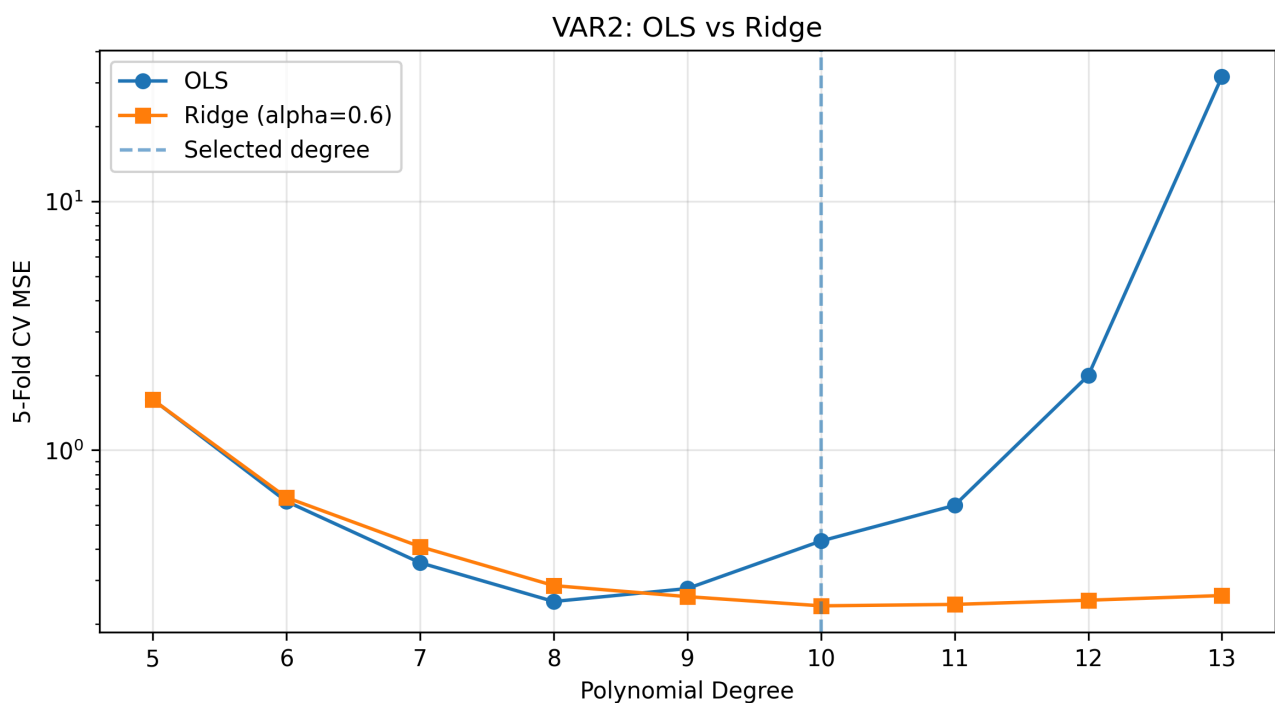


Figure 2: Comparison of OLS and Ridge regression across polynomial degrees using 5-fold cross-validation.

OLS performed best around degree 8, but its validation error increased sharply at higher degrees. Ridge reduced this instability and allowed higher-degree models to perform well.

3.3 Final Model

- **Model:** Ridge regression
- **Polynomial degree:** 10
- **Regularization strength:** $\alpha = 0.6$
- **Selected features:** x_1, x_2, x_3

Across three additional random seeds (7, 21 and 123), the selected model achieved an average CV MSE of **0.236946**, slightly lower than the degree-12 Ridge model (0.237494).

Degree 10 was therefore preferred because it gave comparable performance with fewer polynomial terms.

4. Conclusion

Polynomial regression models were developed for both datasets using OLS and Ridge regression. Different degrees, feature combinations and regularization strengths were compared using 5-fold cross-validation.

Ridge regression gave better validation results for both problems. The final models selected were:

Parameter	Var1	Var2
Model	Ridge	Ridge
Polynomial degree	5	10
Alpha	15	0.6
Features	All 6	All 3
Mean CV MSE (seeds 7, 21, 123)	0.523408	0.236946
Mean CV R ² (seeds 7, 21, 123)	0.947907	0.995131

Regularization was particularly useful for Var1, where it reduced the validation error considerably. For Var2, degree 10 was preferred over degree 12 because their results were very close, and degree 10 uses fewer polynomial terms.

Both final models were trained on all 1,000 training samples, and predictions were generated for the corresponding test datasets.